CS1231S Midterms Cheatsheet

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1. Proofs

Number System Sets

$\mathbb{N}$: Natural numbers $\{0, 1, 2, ...\}$ ($\mathbb{Z}_{\geq 0}$)
$\mathbb{Z}$: Integers $\{3, -100, 3^5, ...\}$
$\mathbb{Q}$: Rational Numbers $\{\frac{1}{2}, -23, 8.6, \frac{-37}{5}, ...\}$
$\mathbb{R}$: Real Numbers $\{-1, \pi, \sqrt{2}, 4.5, ...\}$
** 0 is neither negative or positive*
** $\mathbb{Z}^{+/-}$ set of all positive or negative integers*

Terminologies

Axiom/Postulate: A statement assumed to be true w/o proof. Basic building blocks of theorems
Theorem: A mathematical statement proved using rigorous mathematical reasoning. Major / important result.
Lemma: A small theorem minor result to help prove a theorem.
Corollary: Simple deduction from Theorem.
Conjecture: A statement believed to be true, but no proof yet.

Basic Properties of Integers

Closure: Closed under $+/\times/-$
$x + y \in \mathbb{Z}$ and $xy \in \mathbb{Z}$
Commutativity: Commutative under $+/\times$
$x + y = y + x$ and $xy = yx$
Associativity: Associative under $+/\times$
$x + y + z = (x + y) + z = x + (y + z)$ and $xyz = (xy)z = x(yz)$
Distributivity:
$x(y + z) = xy + xz$
Trichotomy: Exactly one of the following is true
$x = y$ or $x < y$ or $x > y$

Definitions of Number

Even and Odd Integers

Let $n \in \mathbb{Z}$
n is even $\leftrightarrow \exists k \in \mathbb{Z}\ n = 2k$
n is odd $\leftrightarrow \exists k \in \mathbb{Z}\ n = 2k + 1$s

Divisibility

Let $d, n \in \mathbb{Z}\ \wedge d \neq 0$
$d|n \leftrightarrow \exists k \in \mathbb{Z}\ n = dk$
** d divides n*

Prime / Composite

Prime:
$(n > 1) \wedge \forall r, s \in \mathbb{Z}^+, (n = rs \rightarrow (r = 1 \wedge s = n) \vee (r = n \wedge s = 1))$
Composite:
$(n > 1) \exists r, s \in \mathbb{Z}^+ (n = rs \wedge (1 < r < n) \wedge (1 < s < n))$

Congruence

Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$:
$a \equiv b (mod n) \leftrightarrow n|(a - b)$
** forms equivalence relation*

Rationality / Lowest Term Fraction

r is rational $\leftrightarrow \exists a, b \in \mathbb{Z}\ r = \frac{a}{b} \wedge b \neq 0$
** it is lowest term fraction if largest integer that divides $a \wedge b$ in $\frac{a}{b}$ is 1 (prime numbers)*

Methods of Proof

1. **Proof by Construction**: usually proves an existential statement by explicitly finding / constructing a value that satisfies the properties stated.
2. **Proof by Counter Example**: Finding a specific example that shows statement is not always true.
3. **Proof by Exhaustion**: Prove by showing each cases is true. Suitable for finite cases.
4. **Proof by Cases**: Splitting proof into respective cases and try to show the conclusion accordingly
5. **Proof by Deduction**: Form of direct proof usually makes use of definitions. Good for universal statements and infinite cases.
6. **Proof by Contradiction**: Negating the original statement and trying to arrive at a contradiction making the negation to be false.
7. **Proof by Contraposition**: Getting the contrapositive of a statement and direct proof the contrapositive. Might be easier to do this way sometimes.

2. Compound Statements

Definition 2.1.1 (Statement)
A statement / proposition is a sentence that is true or false, but not both
Definition 2.1.2 (Negation)
If p is a statement variable, negation is "not p ($\sim p$)"
Definition 2.1.3 (Conjunction)
"p and q ($p \wedge q$)"
Definition 2.1.4 (Disjunction)
"p or q ($p \vee q$)"
** $\sim$ performed first, then $\wedge$, $\vee$ coequal in order of operation*
Definition 2.1.5 (Statement / Propositional Form) A statement / propositional form is an expression of statement variables and logical connectives that becomes a statement when actual statements are substituted
Definition 2.1.6 (Logical Equivalence)
$P \equiv Q \leftrightarrow$ both have identical truth values
** Use truth table / counter example to prove non-equivalence*
Definition 2.1.7 (Tautology)
Statement that is always true (tautological statement)
Definition 2.1.8 (Contradiction)
Statement that is always false (contradictory statement)

Conditional Statements

Definition 2.2.1 (Conditional)
"if p then q" or "p implies q"
$p\ (hypo/antedecent) \rightarrow q\ (conclusion/consequent)$
** only false when p is true and q is false*
** performed last with $\sim, \wedge, \vee$*
Vacuously true: hypothesis is false
Implication: $\sim p \vee q$
Negation: $p \wedge \sim q$
Definition 2.2.2 (Contrapositive)
"$p \rightarrow q \equiv \sim q \rightarrow \sim p$"
Definition 2.2.3 (Converse)
"$q \rightarrow p$"
Definition 2.2.4 (Inverse)
"$\sim p \rightarrow \sim q$"
** Inverse $\equiv$ Converse*
Definition 2.2.5 (Only If)
p can only take place only if q takes place. "$\sim q \rightarrow \sim p$"
Definition 2.2.6 (Biconditional $\leftrightarrow$)
"$p \leftrightarrow q$" implication but both ways. True when p and q has same value false otherwise.
** Proving it needs to show both direction (Proof by cases $\rightarrow$ and $\leftarrow$)*
** Coequal with Conditional operator*
Definition 2.2.7 (Necessary / Sufficient)
"$p \rightarrow q$" means p is sufficient to guarantee occurrence of q.
"$\sim p \rightarrow \sim q$" means if p does not occur, q does not either. Necessary for p to occur when q occurs

Arguments

Definition 2.3.1 (Argument)

$p \rightarrow q$	(premise)
p	(premise)
$\bullet\ q$	(conclusion)

Valid: All premises (critical rows) and conclusion are true.
Syllogism: Argument form consisting of two premises and a conclusion.

Rules of inference

Modus Ponens

$$\begin{array}{lr} p \to q & \text{(premise)} \\ p & \text{(premise)} \\ \bullet\ q & \text{(conclusion)} \end{array}$$

Modus Tollens

$$\begin{array}{lr} p \to q & \text{(premise)} \\ \sim q & \text{(premise)} \\ \bullet\ \sim p & \text{(conclusion)} \end{array}$$

Generalization

$$\begin{array}{lr} p & \text{(premise)} \\ \bullet\ p \vee q & \text{(conclusion)} \end{array}$$

Specialization

$$\begin{array}{lr} p \wedge q & \text{(premise)} \\ \bullet\ p & \text{(conclusion)} \end{array}$$

Elimination

$$\begin{array}{lr} p \vee q & \text{(premise)} \\ \sim q & \text{(premise)} \\ \bullet\ p & \text{(conclusion)} \end{array}$$

Transitivity

$$\begin{array}{lr} p \to q & \text{(premise)} \\ q \to r & \text{(premise)} \\ \bullet\ p \to r & \text{(conclusion)} \end{array}$$

Div into Cases

$$\begin{array}{lr} p \vee q & \text{(premise)} \\ p \to r & \text{(premise)} \\ q \to r & \text{(premise)} \\ \bullet\ r & \text{(conclusion)} \end{array}$$

Contradiction

$$\begin{array}{lr} \sim p \to false & \text{(premise)} \\ \bullet\ p & \text{(conclusion)} \end{array}$$

Fallacies

1. Converse
2. Inverse
3. False Premise

Definition 2.3.2 (Sound / Unsound Arguments)
Sound $\leftrightarrow$ if it is valid and premises are true
Unsound $\leftrightarrow \sim$ Sound

3. Quantified Statement

Definition 3.1.1 (Predicate): is a sentence that contains a finite number of variables that becomes a statement when values are substituted
Definition 3.1.2 (Truth Set): set of all elements in Domain of P(x) such that P(x) is true
Definition 3.1.3 (Universal Statement $\forall$):
"$\forall x \in D, Q(x)$"
** true $\leftrightarrow$ $Q(x) \equiv$ True for every x in D*
** false $\leftrightarrow$ $Q(x) \equiv$ False for at least one x in D (counter-example)*
Definition 3.1.4 (Existential Statement $\exists$):
"$\exists x \in D, Q(x)$"
** true $\leftrightarrow$ $Q(x) \equiv$ True for at least one x in D*
** false $\leftrightarrow$ $Q(x) \equiv$ False for all x in D (counter-example)*
** $\exists!$: there exists a unique / one and only one*
Theorem 3.2.1 Negation of a Universal Statement:
"$\sim (\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x)$"
Theorem 3.2.2 Negation of a Existential Statement:
"$\sim (\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x)$"
Definition 3.2.1 (Contrapositive, converse, inverse):
Given "$\forall x \in D(P(x) \to Q(x))$"

1. Contrapositive: "$\forall x \in D(\sim Q(x) \to \sim P(x))$"
2. Converse: "$\forall x \in D(Q(x) \to P(x))$"
3. Inverse: "$\forall x \in D(\sim P(x) \to \sim Q(x))$"

Definition 3.2.2 (Sufficient / necessary, Only if):

1. **Sufficient**: "$\forall x(r(x) \to s(x))$" r(x) is sufficient condition for s(x)
2. **Necessary**: "$\forall x(\sim r(x) \to \sim s(x))$" r(x) is a necessary condition for s(x)
3. **Only if**: "$\forall x(\sim s(x) \to \sim r(x))$" r(x) only if s(x)

Rules of Inference Quantifiers

Universal Modus Ponens

$$\begin{array}{lr} \forall x(P(x) \to Q(x)) & \text{(premise)} \\ P(a) & \text{(for particular a — premise)} \\ \bullet\ Q(a) & \text{(conclusion)} \end{array}$$

Universal Modus Tollens

$$\begin{array}{lr} \forall x(P(x) \to Q(x)) & \text{(premise)} \\ \sim Q(a) & \text{(for particular a — premise)} \\ \bullet\ \sim P(a) & \text{(conclusion)} \end{array}$$

Universal Transitivity

$$\begin{array}{lr} \forall x(P(x) \to Q(x)) & \text{(premise)} \\ \forall x(Q(x) \to R(x)) & \text{(premise)} \\ \bullet\ \forall x(P(x) \to R(x)) & \text{(conclusion)} \end{array}$$

5. Set Theory

Set notations:

- **Set-Roster Notation**: $\{1, 2, 3, ...\}$
- **Set-Builder Notation**: $\{x \in U : P(x)\}$
- **Replacement Notation**: $\{t(x) : x \in U\}$

** : or |*
** Order and duplicates do not matter*
Definitions 5.1.2 (Subset, Proper Susbset Empty Set, Singleton):
Subset $\subseteq$: $A \subseteq B \leftrightarrow \forall x(x \in A \to x \in B)$
Proper Subset $\subsetneq$: $A \subsetneq B \leftrightarrow A \subseteq B \wedge A \neq B$
Not a subset $\not\subseteq$: $A \not\subset B \leftrightarrow \exists x(x \in A \wedge x \notin B)$
Theorem 6.2.4 (Empty set subset of every set)
$\emptyset \subseteq A$ (A is any set)
Singleton: a set with single element
Definition 5.1.3 (Ordered Pair)
Form of (x, y). Equality: $(a, b) = (c, d) \leftrightarrow (a = c) \wedge (b = d)$
Definition 5.1.4 (Cartesian Product)
$A \times B = \{(a, b) : a \in A \wedge b \in B\}$
Definition 5.1.5 (Set Equality)
$A = B \leftrightarrow A \subseteq B \wedge B \subseteq A$
$A = B \leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$
** provide $A \subseteq B$ then prove $B \subseteq A$*
Union $\cup$
$A \cup B = \{x \in U : x \in A \vee x \in B\}$
Intersection $\cap$
$A \cap B = \{x \in U : x \in A \wedge x \in B\}$
Difference / Relative Complement $- \setminus$
$B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$
Complement $\bar{A}$
$\bar{A} = \{x \in U : x \notin A\}$
Definition 5.1.8 (Partition of Sets)
For all $i, j, ... = 1, 2, 3$
$A_i \cap A_j = \emptyset | i \neq j$
Theorem 4.4.1 (The Quotient-Remainder Theorem)
For any integer n and positive integer d, there is a unique q and r such that:
$n = dq + r \mid (0 \leq r < d)$
Definition 5.1.9 (Power Set):
Given $A = \{x, y\}$, $\mathcal{P}(A)$ is the set of all subsets of A

$\mathcal{P}(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$
** 2^n elements for power set*
Theorem 6.3.1 (Power Set No. of Elements)
$|\mathcal{P}(A)| = 2^{|A|}$
Theorem 6.2.1 Some Subset Relations (Property of Sets)

1. Inclusion of Intersection: For all sets A and B
   (a)$A \cap B \subseteq A$ (b) $A \cap B \subseteq B$
2. Inclusion in Union: For all sets A and B
   (a)$A \subseteq A \cup B$ (b)$B \subseteq A \cup B$
3. Transitive Property of Subsets: For all sets A, B and C
   $A \subseteq B \wedge B \subseteq C \rightarrow A \subseteq C$

6. Relations

Relations

Defintion 6.1.1 (Relation R):
$xRy \leftrightarrow (x, y) \in R$
** $R \subseteq A \times B$*
Domain $Dom(R)$**:** $\{a \in A : aRb\ \exists b \in B\}$
Co-Domain $coDom(R)$**:** set B
Range $Range(R)$**:** $\{b \in B : aRb\ \exists a \in A\}$
Definition 6.1.2 (Inverse of a Relation R^{-1}):
Given R from A to B. Inverse:
$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$
Definition 6.1.4 (Composition of Relations):
Given sets A, B and C. $R \subseteq A \times B\ S \subseteq B \times C$:
$\forall x \in A, \forall x \in C(xS \circ Rz \leftrightarrow (\exists \in B(xRy \wedge yRz)))$
** 2nd relation first*
Proposition: Composition is Associative:
$T \circ (S \circ R) = (T \circ S) \circ R = T \circ S \circ R$
Proposition: Inverse of Composition
$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$
Definition: (Relexivity, Symmetry, Transitivity):

1. Reflexive: R is reflexive $\leftrightarrow \forall x \in A(xRX)$
2. Symmetric: R is Symmetric $\leftrightarrow \forall x, y \in A(xRy \rightarrow yRx)$
3. Transitive: R is Transitive
   $\leftrightarrow \forall x, y, z \in A(xRy \wedge yRx \rightarrow xRz)$

Transitive Closure
R^t is a transitive closure when it adds least no. of ordered pairs to ensure transitivity:

1. R^t is transitive
2. $R \subseteq R^t$
3. If S is another transitive relation, $R^t \subseteq S$

Equivalence Relation

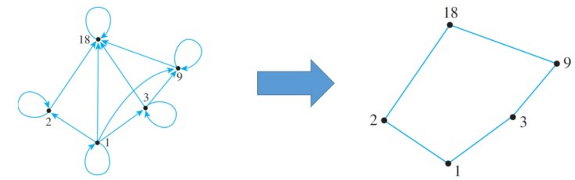
Definition (Partition $\mathcal{C}$):
$\mathcal{C}$ is a partition of set A if the following holds:

1. $\forall S \in \mathcal{C}, (\emptyset \neq S \wedge S \subseteq A)$
   ** S is a subset of A*
2. $\forall x \in A\ \exists S \in \mathcal{C}(x \in S) \wedge \forall x\ \exists S_1, S_2 \in \mathcal{C}(x \in S_1 \wedge x \in S_2 \rightarrow S_1 = S_2)$
   ** Every element in A is in exactly one element of $\mathcal{C}$*
3. Shorter Ver.: $\forall x \in A \exists! S \in \mathcal{C}(x \in S)$

Theorem 8.3.1 (Relation Induced by a Partition):
R induced by partition is reflexive, symmetric and transitives
** same component as relation*
Definition 6.3.2 (Equivalence Relation $\sim$):
R is an equiv relation $\leftrightarrow$ R is reflexive, symmetric, transitive
** Prove by each cases of reflexive, symmetric, transitive*
Definition 6.3.3 (Equivalence Class $[a]_\sim$):
$[a]_\sim = \{x \in A : a \sim x\}$
** The set consisting of elements related to a*
$\forall a \in A(x \in [a]_\sim \leftrightarrow a \sim x)$
Lemma Rel.1 Equivalence Class
$x \sim y \equiv [x] = [y] \equiv [x] \cap [y] \neq \emptyset$
Definition 6.3.5 (Set of Equivalent Classes):
$A/\sim = \{[x]_\sim : x \in A\}$
** set of all equivalence classes*
Theorem 8.3.4 (Partiton Induced By Equiv Relation):
Partition formed by distinct equivalence classes formed by R on set A
Theorem Rel.2 ($A/\sim$ is a partition of A):
Partition formed by distinct equivalence classes formed by R on set A
Proposition $\sim_n$: Congruence-mod n is an equivalence relation on $\mathbb{Z}$ for every $n \in \mathbb{Z}^+$
** Partition into Theorem 4.4.1 Quotient-Remainder Theorem*

Partial Order Relations

Definition 3.4.1 (Antisymmetry):
R is antisymmetric $\leftrightarrow \forall x, y \in A(xRy \wedge yRx \rightarrow x = y)$
Defintion 6.4.2 (Partial Order Relation $\preceq$)
R is a partial order relation $\leftrightarrow$ R is reflexive, antisymmetric, transitive ** e.g* $(\leq / \subseteq)$
Defintion (Partial Order Set (A, R)):
Given partial order relation R on set A, A is the poset
** divides relation is partial order*
Hasse Diagram



Definition Comparability:
$\forall a, b \in A$ (a, b comparable $\leftrightarrow a \leq b \vee b \leq a$)
** noncomparable is the opposite* **Definition 6.4.5 (Maximal/Minimal/Largest/Smallest):**
Given a poset (A, R):

1. c is maximal: $\forall x \in A(c \preceq x \rightarrow c = x)$
2. c is minimal: $\forall x \in A(x \preceq c \rightarrow c = c)$
3. c is largest / greatest: $\forall x \in A(x \preceq c)$
4. c is smallest / least: $\forall x \in A(c \preceq x)$

Proposition: Any smallest element is minimal (largest is maximal)
Definition (Total Order Relation):
R is a partial order $\wedge\ \forall x, y \in A(xRy \vee yRx)$
Definition (Linearization):
$\forall x, y \in A(x \preceq y \rightarrow x \preceq^* y)$
** deriving one total order from partial order*
Kahn's Algorithm:

1. Set $A_0 :=$ A and $i := 0$
2. Repeat until $A_i = \emptyset$
   (a) find a minimal element $c_i \in A_i$ w.r.t $\preceq$
   (b) $A_{i+1} = A_i \setminus \{c_i\}$
   (c) $i := i + 1$
3. Output: Linearization of partial order

Definition (Well-Ordered Set):
$\forall S \in \mathcal{P}(A), S \neq \emptyset \rightarrow (\exists x \in S\ \forall y \in S(x \preceq y)))$
** must be a total order relation*

Proven, Theorems etc.

Numbers

Product of two consecutive odd numbers is always odd
The product of two irrational numbers is not always irrational
Difference of two consecutive squares is always odd
Theorem 4.2.1: Every integer is a rational number
Theorem 4.2.2: The sum of any two rational numbers is rational
Corollary 4.2.3: The double of a rational number is rational
Theorem 4.3.1: A positive divisor of a positive integer
$\forall a, b \in \mathbb{Z}^+ (a|b \to a \leq b)$
Theorem 4.3.2: Divisors of 1 are 1 and -1
Theorem 4.3.3 Transitivity of Divisibility
Theorem 4.6.1: There is no greatest integer
Theorem 4.7.1: $\sqrt{2}$ is irrational
Proposition 4.6.4: For all integers n, if n^2 is even then n is even
Theorem 4.4.1 (The Quotient-Remainder Theorem)
For any integer n and positive integer d, there is a unique q and r such that:
$n = dq + r \mid (0 \leq r < d)$

Quantifiers

Theorem 3.2.1 Negation of a Universal Statement:
"$\sim (\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x)$"
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$A \subseteq B \wedge B \subseteq C \to A \subseteq C$

Relations

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